

USAMA IMRAN CHEEMA

Linux Systems Administrator & Cloud Infrastructure Engineer

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EXECUTIVE SUMMARY

Proactive **Linux Systems Administrator & Cloud Infrastructure Engineer** with **3+ years of intensive hands-on experience** in low-level Linux administration, POSIX systems engineering, server security hardening, and hybrid cloud operations (Azure & AWS). Proven track record of engineering high-throughput systems software in C/POSIX (mmap, atomic bitfields, lockless concurrency), developing automated security auditing toolchains, and maintaining 24/7 virtualized server clusters using **QEMU/KVM**. Skilled in deep networking diagnostics (TCP/IP, DNS, Nginx reverse proxy, firewalls) and 24/7 uptime monitoring. Dedicated to building, automating, and securing high-reliability production infrastructure.

CORE TECHNICAL SKILLS

Operating Systems & Virtualization:	Linux (Ubuntu Server, Debian, RHEL/Rocky, Arch Linux), QEMU/KVM, Libvirt, Proxmox, Docker, Kernel/POSIX APIs
Server Administration & Ops:	Systemd daemons, SSH Key Hardening, User/Group RBAC, Log Diagnostics (journalctl, dmesg), top, ss, df, cron
Cloud Platforms:	Microsoft Azure (Linux VMs, VNets, Subnets, NSGs, Blob Storage), AWS (EC2, VPC, S3, IAM, SigV4 Auth)
Networking & Security:	TCP/IP, DNS, DHCP, Nginx Reverse Proxy, SSL/TLS (Let's Encrypt), UFW/Iptables, Fail2ban, Threat Intelligence Auditing
Systems Scripting & Tools:	C / POSIX Systems APIs (posix_fallocate, pwrite, mmap), Python, Bash Automation, Go, Git/GitHub, Cloud-init

FEATURED SYSTEMS & INFRASTRUCTURE PROJECTS

down — High-Performance Concurrency & POSIX Transfer Engine

C11, POSIX, mmap, Kernel Primitives, AWS SigV4

- Engineered a lockless, multi-threaded transfer engine in C/POSIX using posix_fallocate, atomic pwrite, and mmap.
- Implemented crash recovery with an atomic bitfield tracking 512 KB chunks; designed a dynamic work-stealing scheduler.
- Integrated OpenSSL cryptographic hashing (SHA-256, BLAKE2) and AWS SigV4 authentication for cloud storage.

containr — Linux Container Runtime from Scratch

Go, Linux Namespaces, cgroups v2, Syscalls, Isolation

- Architected a lightweight container engine in Go utilizing CLONE_NEWPID, NEWUTS, NEWNS, and NEWIPC namespaces.
- Enforced kernel **cgroups v2** resource constraints (memory.max, pids.max, cpu.max) and private mount propagation (MS_REC | MS_PRIVATE) with chroot rootfs isolation.

prom-linux-exporter — Zero-Dependency Linux Telemetry Exporter

Go, Prometheus, Linux Kernel /proc, Docker, Grafana

- Built a high-performance Go telemetry daemon scraping real-time metrics directly from /proc/{stat,meminfo,loadavg,diskstats,net/dev}.
- Exposed standard Prometheus metrics (:9100/metrics) with multi-stage Docker containerization and automated Grafana dashboards.

Enterprise Homelab & Multi-Node Linux Virtualization

QEMU/KVM, Bare-Metal Linux, Systemd, UFW, Nginx

- Architected a 24/7 self-hosted server cluster on bare-metal Linux and QEMU/KVM virtual machines simulating enterprise networks.
- Configured local DNS, DHCP, Nginx reverse proxy with SSL/TLS renewal, strict UFW firewalls, and Fail2ban intrusion protection.

EDUCATION & BACKGROUND

Self-Directed Computer Systems Curriculum

2022 – Present | OS Architecture, Kernel Internals, Networking, Cloud

Undergraduate University Studies

4 Semesters Completed | Analytical Reasoning & Scientific Research

CERTIFICATIONS (IN PROGRESS)

Red Hat Certified System Administrator (RHCSA EX200)

In Active Preparation – Live Lab Testing Track

Microsoft Certified: Azure Administrator (AZ-104)

In Active Preparation – Cloud Compute & VNets